6.1.2 Text Dataset

Large Movie Review Dataset Large Movie Review [38] is a binary dataset for sentiment classification (positive or negative) consisting of 25,000 highly polar movie reviews for training, and 25,000 reviews for testing.

6.2 Proofs

Theorem 1 *The gradient propagation of a stochastic unit* h *based on a deterministic function* g *with inputs* $\boldsymbol{x}$ *(a vector containing outputs from other neurons), internal parameters* $\boldsymbol{\phi}$ *(weights and bias) and noise* z *is possible, if* $g(\boldsymbol{x}, \boldsymbol{\phi}, z)$ *has non-zero gradients with respect to* $\boldsymbol{x}$ *and* $\boldsymbol{\phi}$*. [24]*

$$h = g(\boldsymbol{x}, \boldsymbol{\phi}, z) \tag{8}$$

Assume a network has two layers with a unit for each layer, the distribution of the second layer's output y_2 differs depending on the first and second layer's weights (w_1 and w_2) and sigmas (σ_1 and σ_2) as:

$$\begin{aligned} y_2 &= \mu\left[w_2\mu(w_1x) + w_2\sigma_1\epsilon\right] + \sigma_2\epsilon \\ &= \begin{cases} \mu\left[w_2\mu(w_1x))\right] + w_2\sigma_1\epsilon + \sigma_2\epsilon & \text{if } w_2\mu(w_1x) + w_2\sigma_1\epsilon > 0, \\ \sigma_2\epsilon & \text{otherwise,} \end{cases} \\ &\sim \begin{cases} N(\mu\left[w_2\mu(w_1x))\right], (w_2\sigma_1)^2 + \sigma_2^2) \\ N(0, \sigma_2^2). \end{cases} \end{aligned} \tag{9}$$

Incidentally, as shown in Figure 1 (c) in the main paper, a small noise variance tends to be learned in the final layer to make the network output stable (see Section 4.3 in the paper for a quantitative evaluation).

Using Eq. (8) from Theorem 1, assume g is noise injection function that depends on noise z, and some differentiable transformations $\boldsymbol{d}$ over inputs $\boldsymbol{x}$ and model internal parameters $\boldsymbol{\phi}$. We can derive the output, h as:

$$h = g(\boldsymbol{d}, \boldsymbol{z}) \tag{10}$$

If we use Eq. (10) for another noise addition methods like dropout [43] or masking the noise in denoising auto-encoders [44], we can infer z as noise multiplied just after a non-linearity is induced in a neuron. In the case of ProbAct, we sample from Gaussian noise and add it while computing h. Or we can say, we add a noise to the pre-activation, which is used as an input to the next layer. In doing so, self regularization behaviour is induced in the network.

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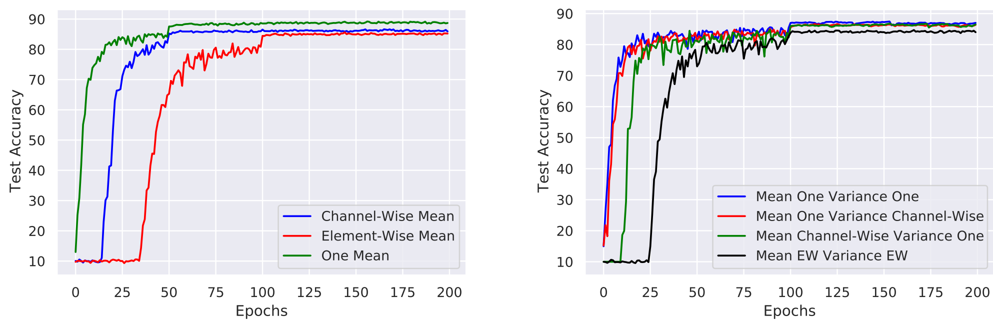


(a) (b)

Figure 6: (a) Comparison of different mean settings with fixed $\sigma = 1$ for CIFAR-10 datasets. Channel-Wise mean denotes a constant learnable mean within the channels, element-wise mean denotes learnable mean for each parameter, while one mean denotes a single learnable mean for entire dataset. It is worth noting that element-wise mean takes a lot more time to converge compared to one mean or channel-wise mean. This shows that using single learnable mean uses fewer parameters and converges faster.(b) denotes the different variance settings for CIFAR-10 dataset. On contrary to mean settings, there is no clear convergence time difference between single variance and channel-wise variance. Single variance is preferred due to the usage of fewer parameters.Training element-wise mean and element-wise variance (denoted as EW) leads to too many learnable parameters and takes a long time to converge. Training time is high for such a solution and is only preferred when overfitting is an issue.

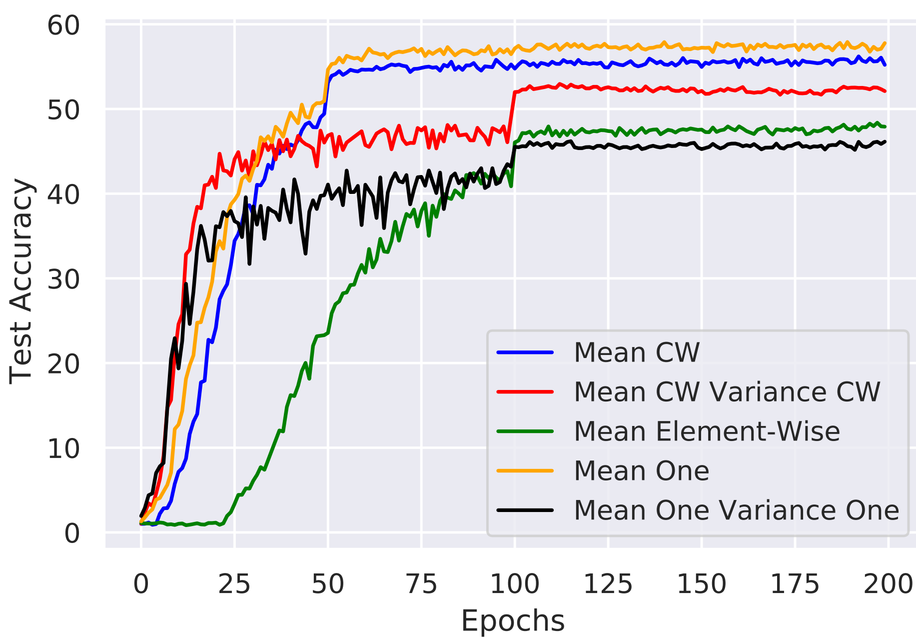
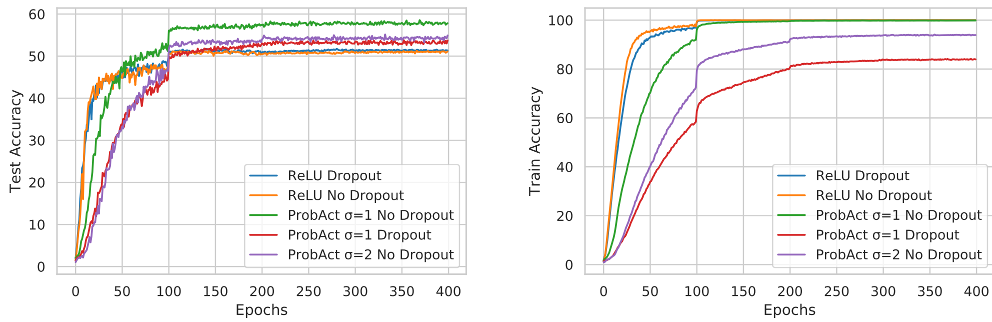


Figure 7: Different mean and variance setting of ProbAct for CIFAR-100 dataset. It is worth noting that single learnable mean is equally effective compared to channel-wise learnable mean but with fewer parameters. Element-wise mean accuracy is lesser than the others but it is more susceptible to overfitting as stated in the paper.



(a) (b)

Figure 8: (a) and (b) shows the train and test accuracy comparison between ReLU and ProbAct layer with/without dropout layers on CIFAR-100 dataset. With $\sigma = 2$, ProbAct achieves similar test performance to ReLU activation layer with dropout with less overfitting as shown from the training curve. Adding a dropout layer further improves the generalization capabilities, showing the built-in regularization nature of ProbAct.

Figure 9: Comparison of predictive test accuracy and confidence score between ReLU (above) and ProbAct (below) on a VGG-16 architecture on CIFAR-10 dataset with 0.05 Gaussian noise added to test samples.

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